

Applications of Differential Calculus (Related Rates)

Name
Date
Course

Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c) = 0$ "c" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n) = nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)] = cf'(x)$ "c" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

The Chain Rule viii) If $h(x) = f(g(x))$, then

$$h'(x) = f'(g(x))g'(x)$$

Rule for Exponential Functions ix) $\frac{d}{dx}(a^x) = a^x \ln a$, "a" is a constant and $a > 0$.

Rule for base "e" Exponential Functions x) $\frac{d}{dx}(e^x) = e^x$

Rule for Logarithmic Functions xi) $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$, "a" is a constant and $a > 0$.

Rule for the Natural Logarithm Function xii) $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Rule for Composite Natural Logarithm Functions xiii) $\frac{d}{dx}\{\ln[g(x)]\} = \frac{g'(x)}{g(x)}$

Trigonometric Differentiation Formulas

I) $\frac{d}{dx}(\sin x) = \cos x$ II) $\frac{d}{dx}(\cos x) = -\sin x$

III) $\frac{d}{dx}(\csc x) = -\csc x \cot x$ IV) $\frac{d}{dx}(\sec x) = \sec x \tan x$

V) $\frac{d}{dx}(\tan x) = \sec^2 x$ VI) $\frac{d}{dx}(\cot x) = -\csc^2 x$

Procedure for Solving Related Rates Problems

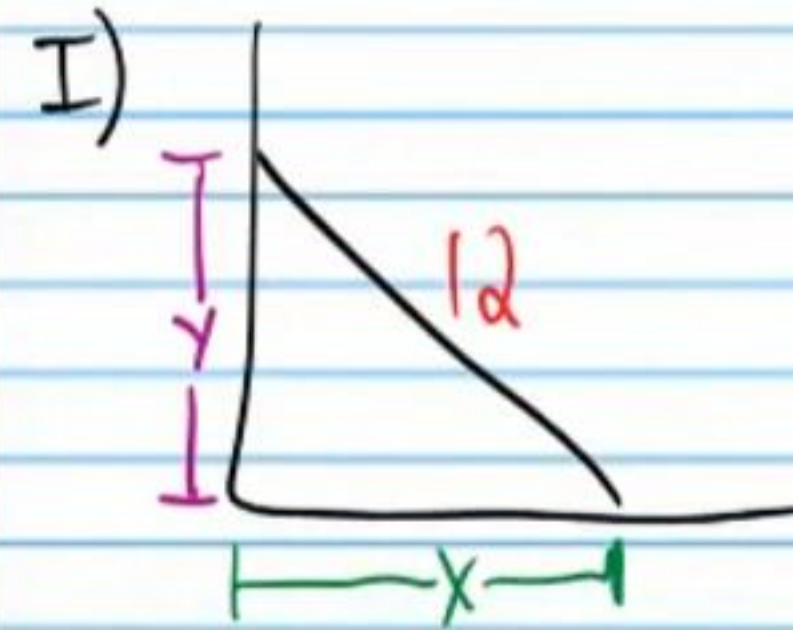
I) Draw a picture and label variables and constants. Some problems may require only a simple drawing.

II) Find an equation relating the variables in question. In many cases, some work must be done to write the equation using only two variables. In some rare cases, the problem can be solved when there is more than two variables in the equation.

III) Differentiate both sides of the equation, remembering to use the chain rule given that each variable is dependent (usually on time, sometimes on other things).

IV) Plug in known values into the equation (some additional math may be required to find all necessary values), and then solve for the unknown value.

- ① A 12 ft ladder rests against a wall. If the ladder slides away from the wall at 1.5 ft/s, how fast is the top of the ladder moving down the wall when the bottom of the ladder is 7 ft from the wall?



II) $a^2 + b^2 = c^2$ III) $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

$$x^2 + y^2 = 12^2$$

$$x^2 + y^2 = 144$$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

$$(7)(1.5) + \sqrt{95} \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\frac{10.5}{\sqrt{95}}$$

$$\boxed{\frac{dy}{dt} = -1.077 \text{ ft/s}}$$

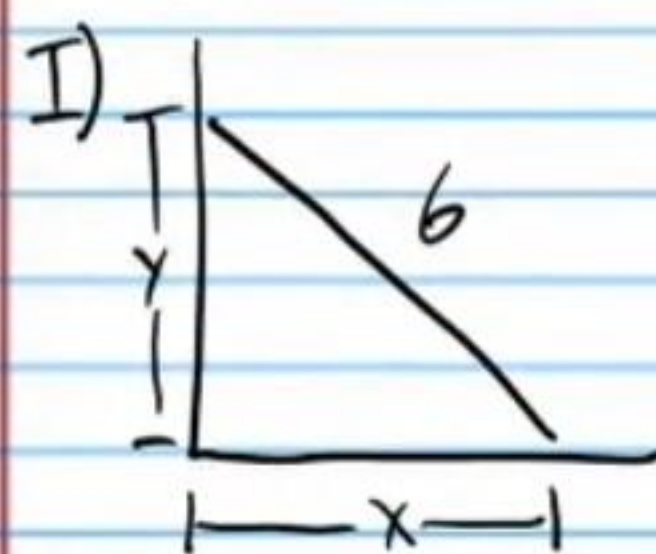
$$x^2 + y^2 = 144$$

$$7^2 + y^2 = 144$$

$$y = \pm \sqrt{95}$$

$$y = \sqrt{95}$$

② A 6 m ladder rests against a wall. If the top of the ladder slides down the wall at 1 m/s, how fast is the bottom of the ladder moving away from the wall when the top of the ladder is 3 m from the ground?



II) $x^2 + y^2 = 6^2$ III) $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$

$x^2 + y^2 = 36$

$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

$3\sqrt{3} \frac{dx}{dt} + 3(-1) = 0$

$\frac{dx}{dt} = \frac{3}{3\sqrt{3}}$

$\frac{dx}{dt} = \frac{\sqrt{3}}{3} \text{ m/s}$

$x^2 + y^2 = 36$

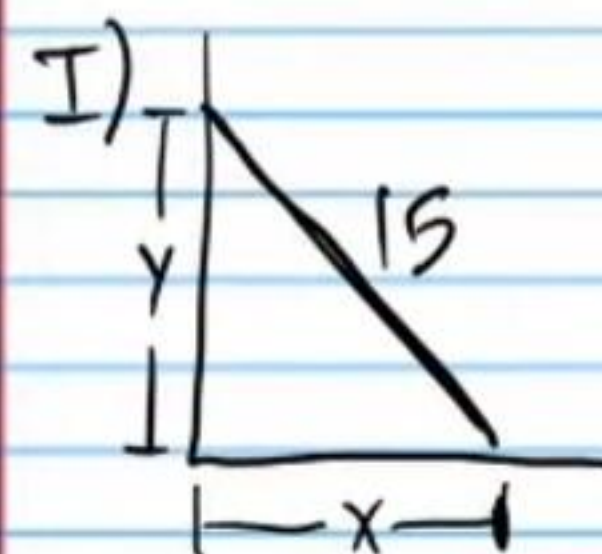
$x^2 + 3^2 = 36$

$x^2 = 27$

$x = \pm 3\sqrt{3}$

$x = 3\sqrt{3}$

③ A 15 ft ladder rests against a wall. If the ladder slides away from the wall at 2 ft/s, how fast is the top of the ladder moving down the wall when the top of the ladder is 11 ft from the ground?



II) $x^2 + y^2 = 15^2$ III) $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$

$x^2 + y^2 = 225$

$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

$(2\sqrt{26})2 + 11 \frac{dy}{dt} = 0$

$\frac{dy}{dt} = -\frac{4\sqrt{26}}{11} \text{ ft/s}$

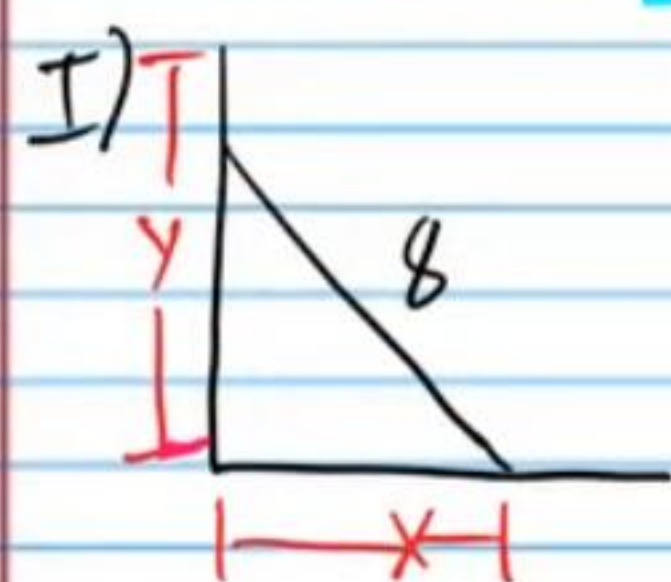
$x^2 + y^2 = 225$

$x^2 + 11^2 = 225$

$x^2 = 104$

$x = 2\sqrt{26}$

- ④ An 8 m ladder rests against a wall. If the top of the ladder slides down the wall at 0.6 m/s, how fast is the bottom of the ladder moving away from the wall when the bottom of the ladder is 3 m from the wall?



II) $x^2 + y^2 = 8^2$ III) $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$

$x^2 + y^2 = 64$

$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

$3 \frac{dx}{dt} + \sqrt{55}(-0.6) = 0$

$\frac{dx}{dt} = \frac{0.6\sqrt{55}}{3}$

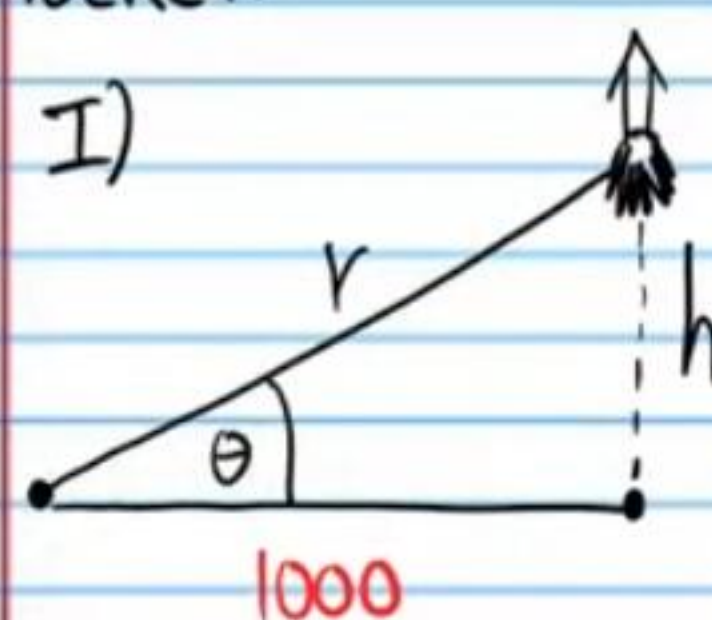
$\frac{dx}{dt} = \frac{\sqrt{55}}{5} \text{ m/s}$

$x^2 + y^2 = 64$

$3^2 + y^2 = 64$

$y = \sqrt{55}$

- ⑤ A camera at ground level films the lift-off of a rocket 1,000 yards from its launch pad. Find the rate of change in the angle of elevation of the camera when the rocket is moving at 15 yd/s and is 1300 yards above the ground. Assume that the camera is continually focused on the rocket.



II) $\tan \theta = \frac{h}{1000}$ III) $\frac{d\theta}{dt} \sec^2 \theta = \frac{1}{1000} \frac{dh}{dt}$

$\tan \theta = \frac{1}{1000} h$

IV) $\frac{d\theta}{dt} \sec^2 \theta = \frac{1}{1000} \frac{dh}{dt}$

$\frac{d\theta}{dt} \left(\frac{r}{1000} \right)^2 = \frac{1}{1000} (15)$

$x^2 + h^2 = r^2$

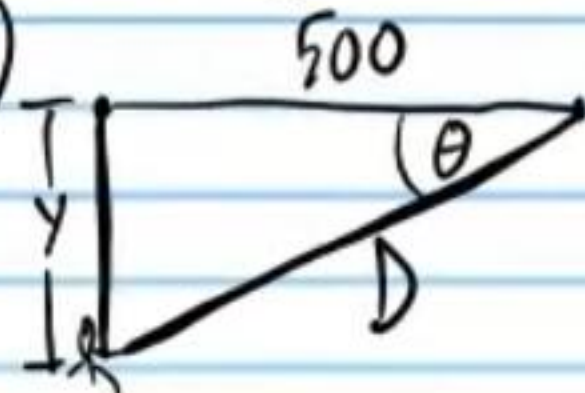
$1000^2 + 1300^2 = r^2$

$r = 1640.1219...$

$\frac{d\theta}{dt} \left(\frac{1640.1219}{1000} \right)^2 = \frac{15}{1000}$

$\frac{d\theta}{dt} = 0.005576 \text{ rad/s}$

⑥ Two helicopters are flying stationary at the same altitude and they are 500 meters apart. The helicopter at the left drops a rescue line to a man on the ground and reels him up while remaining stationary. When the man is 100 m from the helicopter rescuing him, the distance between the helicopter at the right and the man being rescued is changing by 15 m/min. At this moment in time, what is the rate of change of the angle of depression from the helicopter at the right to the man being rescued?

I)  II) $\cos \theta = \frac{500}{D}$ III) $-\frac{d\theta}{dt} \sin \theta = 500 \left(-D^{-2}\right) \frac{dD}{dt}$

$$\cos \theta = 500 D^{-1} \quad -\frac{d\theta}{dt} \sin \theta = -\frac{500}{D^2} \frac{dD}{dt}$$

$$\frac{d\theta}{dt} \sin \theta = \frac{500}{D^2} \frac{dD}{dt}$$

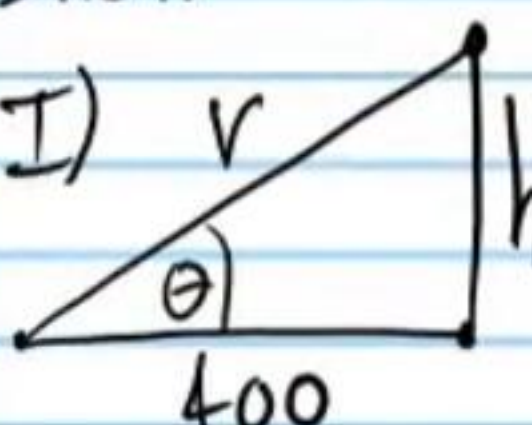
IV) $\frac{d\theta}{dt} \sin \theta = \frac{500}{D^2} \frac{dD}{dt}$

$$\frac{d\theta}{dt} \left(\frac{y}{D}\right) = \frac{500}{(509.9019)^2} (-15)$$

$$\frac{d\theta}{dt} \left(\frac{100}{509.9019}\right) = \frac{500(-15)}{(509.9019)^2}$$

$$\boxed{\frac{d\theta}{dt} = -0.147 \text{ rad/min}}$$

⑦ A cannonball is shot straight up. Find the rate of change of the angle of elevation from a point on the ground to the ball when the altitude of the ball is 600 ft and its speed is 36 ft/s. The point on the ground is 400 ft from where the ball was shot.

I)  II) $\tan \theta = \frac{h}{400}$ III) $\frac{d\theta}{dt} \sec^2 \theta = \frac{1}{400} \frac{dh}{dt}$

$$\tan \theta = \frac{1}{400} h$$

IV) $\frac{d\theta}{dt} \sec^2 \theta = \frac{1}{400} \frac{dh}{dt}$

$$\frac{d\theta}{dt} \left(\frac{r}{400}\right)^2 = \frac{1}{400} (36)$$

$$400^2 + h^2 = r^2$$

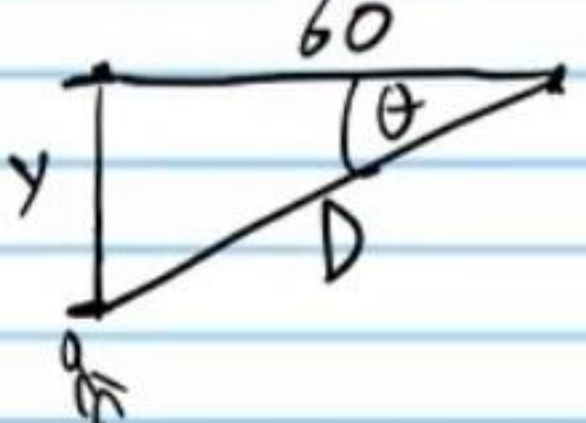
$$400^2 + 600^2 = r^2$$

$$r = 721.11$$

$$\frac{d\theta}{dt} \left(\frac{721.11}{400}\right)^2 = \frac{36}{400}$$

$$\boxed{\frac{d\theta}{dt} = 0.02769 \text{ rad/s}}$$

⑧ Two hot air balloons are flying stationary at the same altitude and they are 60 yards apart. The balloon at the left drops a rescue line to a man on the ground and reels him up while remaining stationary. When the man is 32 yards from the balloon rescuing him, the distance between the balloon at the right and the man being rescued is changing by 20 yd/min. At this moment in time, what is the rate of change of the angle of depression from the balloon at the right to the man being rescued?

I)  II) $\cos \theta = \frac{60}{D}$ III) $-\frac{d\theta}{dt} \sin \theta = \frac{0(D) - 60 \frac{dD}{dt}}{D^2}$

$$\cos \theta = 60 D^{-1}$$

$$-\frac{d\theta}{dt} \sin \theta = \frac{-60 \frac{dD}{dt}}{D^2}$$

$$\frac{d\theta}{dt} \sin \theta = 60 \frac{dD}{dt} D^{-2}$$

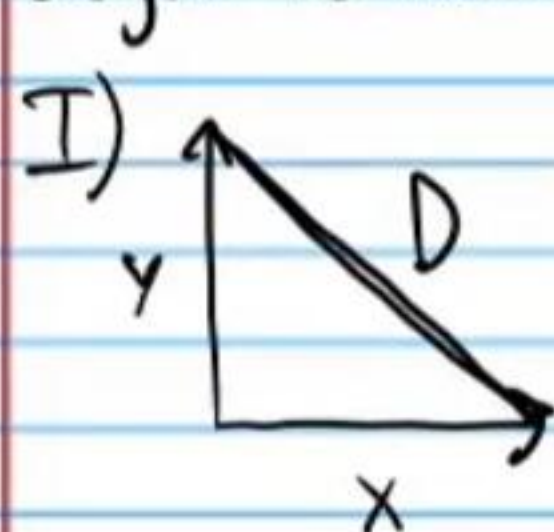
IV) $\frac{d\theta}{dt} \sin \theta = 60 \frac{dD}{dt} D^{-2}$

$$\frac{d\theta}{dt} \left(\frac{y}{D} \right) = 60 \frac{(-20)}{D^2}$$

$$\frac{d\theta}{dt} \frac{32}{68} = \frac{60(-20)}{(68)^2} = \boxed{-0.55 \text{ rad/min}}$$

$$\begin{aligned} 60^2 + y^2 &= D^2 \\ 60^2 + 32^2 &= D^2 \\ D &= 68 \end{aligned}$$

• ⑨ Two cars start to move from the same location. One travels north at 42 mi/h. The other travels east at 30 mi/h. What is the rate of change of the distance between the two cars 3 hours after they begin to move?



II) $x^2 + y^2 = D^2$ III) $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2D \frac{dD}{dt}$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$

$$x = (30 \text{ mi/h})(3 \text{ h}) = 90 \text{ mi}$$

$$y = (42 \text{ mi/h})(3 \text{ h}) = 126 \text{ mi}$$

$$(90)(30) + 126(42) = 154.84185 \frac{dD}{dt}$$

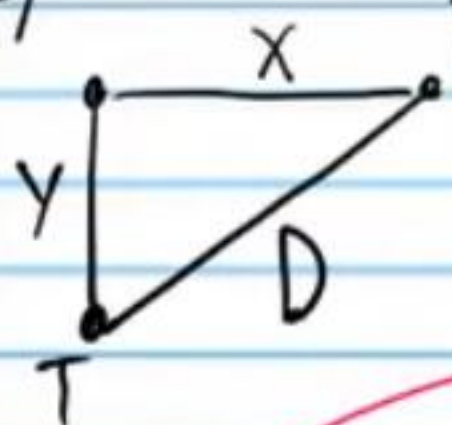
$$x^2 + y^2 = D^2$$

$$90^2 + 126^2 = D^2$$

$$D = 154.84185$$

$$\boxed{\frac{dD}{dt} = 51.61 \text{ mi/h}}$$

⑩ Jim is standing 200 ft east of Tony. They both begin to run at the same time. Jim runs west at 6 ft/s and Tony runs south at 9 ft/s . How fast is the distance between the men changing 8 seconds after they begin to run?

I)  II) $x^2 + y^2 = D^2$ III) $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2D \frac{dD}{dt}$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$ $x = 200 - (6 \text{ ft/s})(8 \text{ s})$

$152(-6) + (72)9 = 168.19036 \frac{dD}{dt}$ $x = 200 - 48$

$x = 152 \text{ ft}$

$$\boxed{\frac{dD}{dt} = -1.57 \text{ ft/s}}$$

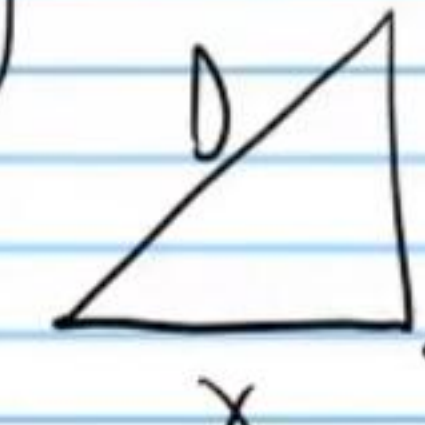
$y = (9 \text{ ft/s})(8 \text{ s}) = 72 \text{ ft}$

$x^2 + y^2 = D^2$

$152^2 + 72^2 = D^2$

$D = 168.190368$

⑪ Two ships approach the same destination. One moves south at 20 km/h and the other moves east at 35 km/h . At what rate is the distance between the two ships changing when the first ship is 2 km from the destination and the second ship is 6 km from the destination?

I)  II) $x^2 + y^2 = D^2$ III) $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2D \frac{dD}{dt}$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$

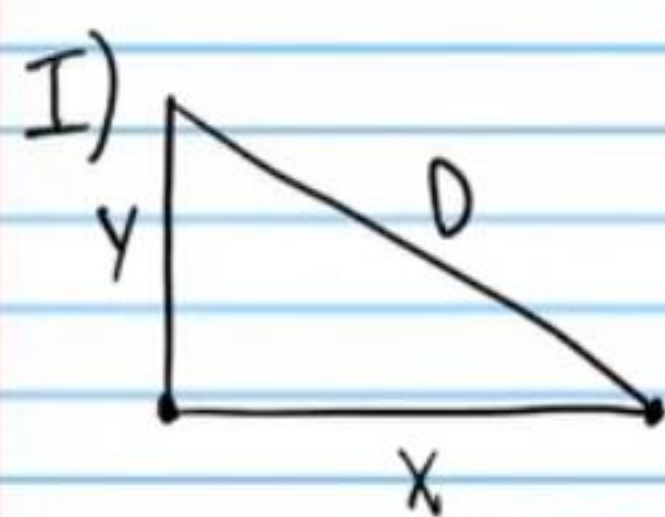
$(6)(-35) + (2)(-20) = 2\sqrt{10} \frac{dD}{dt}$

$x^2 + y^2 = D^2$
 $6^2 + 2^2 = D^2$

$D = 2\sqrt{10}$

$$\boxed{\frac{dD}{dt} = -39.528 \text{ km/h}}$$

⑫ A plane is 2000 miles west of a second plane. The first plane is flying north at 300 mi/h and the second plane is flying west at 250 mi/h. How fast is the distance between the planes changing 5 hours later if the altitudes of the planes are unchanged?



I) $x^2 + y^2 = D^2$ III) $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2D \frac{dD}{dt}$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$$

IV) $x \frac{dx}{dt} + y \frac{dy}{dt} = D \frac{dD}{dt}$

$$750(-250) + 1500(300) = (1677.05) \frac{dD}{dt}$$

$$\frac{dD}{dt} = 156.5 \text{ mi/h}$$

$$x = 2000 - (250 \text{ mi/h})(5 \text{ h})$$

$$x = 2000 - 1250$$

$$x = 750 \text{ mi}$$

$$y = (300 \text{ mi/h})(5 \text{ h})$$

$$y = 1500 \text{ mi}$$

$$x^2 + y^2 = D^2$$

$$750^2 + 1500^2 = D^2$$

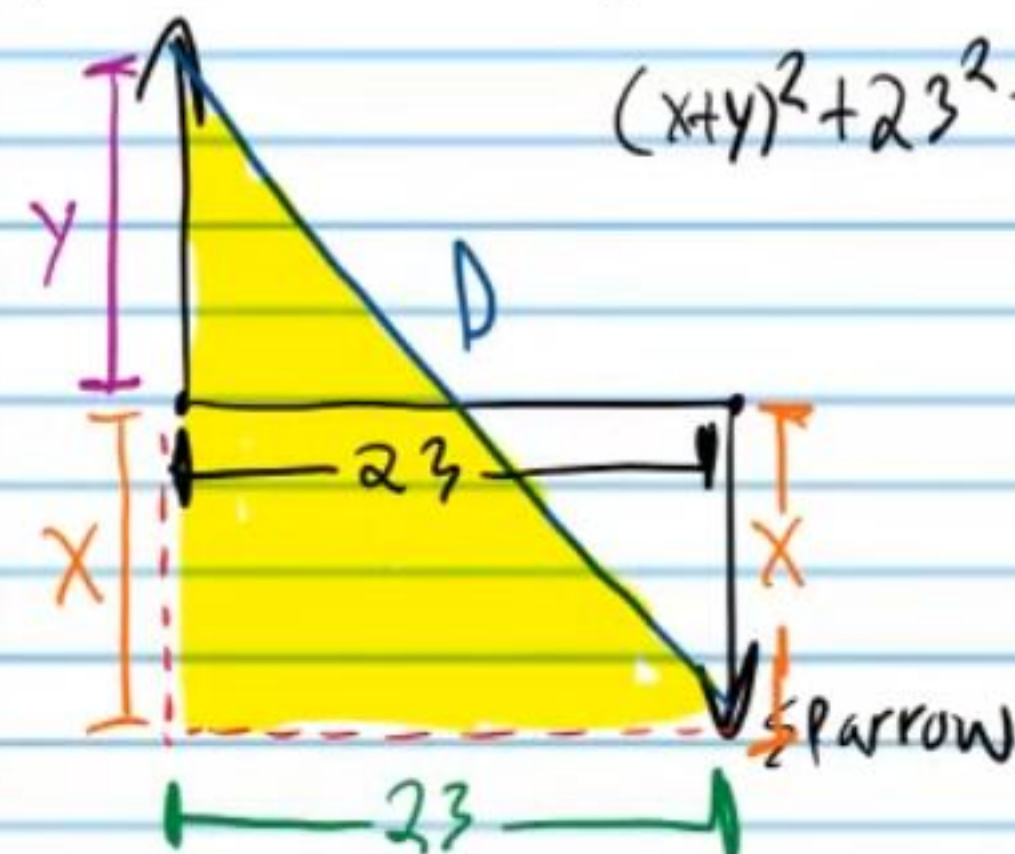
$$D = 1677.05$$

⑬ A seagull is flying 23 miles west of a sparrow at the same altitude. The seagull is flying north at 10 mi/h and the sparrow is flying south at 17 mi/h. How fast is the distance between them changing 20 minutes later assuming that their paths, speeds, and altitudes do not change?

I) Seagull

II) $a^2 + b^2 = c^2$

III) $2(x+y) \left(\frac{dx}{dt} + \frac{dy}{dt} \right) + 0 = 2D \frac{dD}{dt}$



$$(x+y)^2 + 23^2 = D^2$$

$$(x+y) \left(\frac{dx}{dt} + \frac{dy}{dt} \right) = D \frac{dD}{dt}$$

IV) $(x+y) \left(\frac{dx}{dt} + \frac{dy}{dt} \right) = D \frac{dD}{dt}$

$$\left(\frac{17}{3} + \frac{10}{3} \right) (17 + 10) = 24.698 \frac{dD}{dt}$$

$$\frac{dD}{dt} = 9.839 \text{ mi/h}$$

$$20 \text{ minutes} = \frac{1}{3} \text{ hour}$$

$$x = (17 \text{ mi/h}) \left(\frac{1}{3} \text{ h} \right) = \frac{17}{3} \text{ mi}$$

$$y = (10 \text{ mi/h}) \left(\frac{1}{3} \text{ h} \right) = \frac{10}{3} \text{ mi}$$

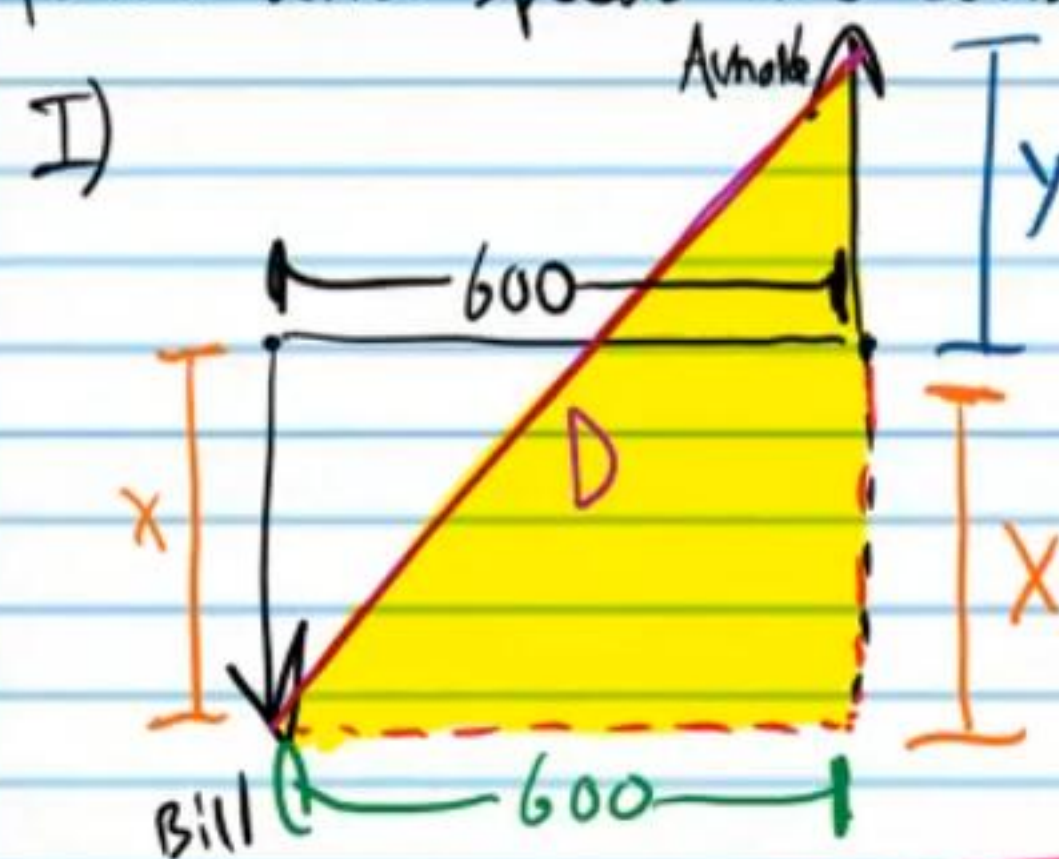
$$(x+y)^2 + 23^2 = D^2$$

$$\left(\frac{17}{3} + \frac{10}{3} \right)^2 + 23^2 = D^2$$

$$9^2 + 23^2 = D^2$$

$$D = 24.698$$

- ⑭ Arnold is standing 600 meters east of Bill. Bill runs south at 30 meters/min and Arnold runs north at 24 meters/min. If they both start running at the same time, what is the rate that the distance between them is changing one hour later assuming that their paths and speeds are constant?



II) $(x+y)^2 + 600^2 = D^2$

III) $2(x+y) \left(\frac{dx}{dt} + \frac{dy}{dt} \right) + 0 = 2D \frac{dD}{dt}$

$(x+y) \left(\frac{dx}{dt} + \frac{dy}{dt} \right) = D \frac{dD}{dt}$

IV) $(x+y) \left(\frac{dx}{dt} + \frac{dy}{dt} \right) = D \frac{dD}{dt}$

1 hour = 60 minutes

$x = (30 \text{ m/min})(60 \text{ min})$
 $= 1800 \text{ m}$

$y = (24 \text{ m/min})(60 \text{ min})$
 $y = 1440 \text{ m}$

$(1800 + 1440)(30 + 24) = 3295.087 \frac{dD}{dt}$

$\frac{dD}{dt} = 53.097 \text{ m/min}$

$(x+y)^2 + 600^2 = D^2$

$(1800 + 1440)^2 + 360,000 = D^2$

$D = 3295.087$